

Strict Global PF₄ for the Riemann Kernel and a Continuous Fourier Separator

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Abstract

Let Φ be the classical Riemann, or de Bruijn–Newman, kernel. We prove that every translation minor of Φ of order at most four is strictly positive at strictly ordered nodes, and an exact origin-confluent calculation gives a negative order-five minor. Thus the exact global Polya-frequency order is four. For the positive theorem, a two-mode argument proves global positivity of the log-curvature quantities q, F_2 and of the fully confluent invariant $\mathcal{C}_4$. The finite verification uses only rational polynomial algebra, Taylor inequalities, and geometric tails. The order-five origin certificate likewise uses exact rational arithmetic. A quotient-Wronskian argument reduces arbitrary fourth-order minors to $\partial_\xi \Psi < 0$. In the increasing curvature coordinate, its numerator is

$$\delta\Lambda \int_p^w W(t) \frac{\mathcal{C}_4(t)}{Q(t)^6 \kappa(t)^2} dt.$$

Here W is the difference of two explicitly normalized triangular CDFs. Its strict positivity follows from a one-crossing density ratio. The transport reductions and analytic sign estimates are given explicitly, with a table of the finite polynomial inequalities used for the three global signs. We then construct an explicit positive even Schwartz kernel which is itself strictly PF₄ but whose entire Fourier transform has nonreal zeros; the latter conclusion follows from a short exact discriminant inequality. Thus continuous PF₃ and PF₄ membership alone do not force Fourier real-rootedness.

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1 Introduction

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is PF_r when every translation-kernel minor

$$\det[f(x_i - y_j)]_{i,j=1}^k$$

is nonnegative for $1 \leq k \leq r$ and increasing node tuples. We call the property strict when the determinant is positive whenever both tuples are strictly increasing. This paper determines the exact finite Polya-frequency order of the Riemann kernel; see [3] for modern background and Schoenberg's characterization. Related finite-order and Riemann-kernel work is placed in Section 2.

Theorem 1.1 (Strict global PF_4). *For every $1 \leq k \leq 4$ and strictly increasing real tuples $x_1 < \dots < x_k$ and $y_1 < \dots < y_k$,*

$$\det[\Phi(x_i - y_j)]_{i,j=1}^k > 0.$$

A parity factorization of the fully confluent derivative matrix at the origin gives a negative order-five limit. Divided differences then transfer that sign to equally spaced translation minors at sufficiently small positive spacing; see Theorem 11.2. Hence Theorem 1.1 has the following exact consequence.

Corollary 1.2. *The exact global Polya-frequency order of Φ is four.*

The exact-order classification and the Fourier question are distinct. At the proved order-four level, the following example shows that finite-order membership alone does not force Fourier real-rootedness, even within the positive even Schwartz class.

Theorem 1.3 (Continuous PF_4 Fourier separator). *There is an explicit positive even Schwartz function f_* which is strict PF_4 , while its entire Fourier transform has nonreal zeros. Consequently continuous PF_3 and continuous PF_4 membership are each insufficient to force Fourier real-rootedness.*

The proof has two one-variable inputs and one exact transport mechanism. First, global inequalities $q > 0$ and $F_2 > 0$ imply strict PF_3 and positivity of a three-point functional Λ . Second, the fully confluent order-four invariant $\mathcal{C}_4$ is globally positive. Exact quotient identities reduce strict PF_4 to the monotonicity of a scalar three-point function Ψ . An increasing curvature coordinate then turns the derivative of Ψ into a positive crossing-kernel integral of $\mathcal{C}_4$.

The finite part of the first input consists of exact rational Bernstein coefficients on fixed algebraic domains, followed by analytic bounds for all remaining theta modes; Appendix B lists the domains, degrees, and minimum coefficients. The transport algebra is proved in the text, and the order-five calculation uses rational enclosures and elementary series bounds. Theorem 1.3, proved in Section 12, supplies an explicit order-four counterexample rather than a finite-order sufficiency threshold.

2 Relation to earlier work

Schoenberg introduced Polya-frequency functions and connected translation minors with bilateral Laplace transforms and variation-diminishing convolution operators [1, 3]. Karlin’s monograph remains the standard systematic account of total positivity and its composition principles [2]; a modern review of PF functions and their preservers is given by Belton, Guillot, Khare, and Putinar [3].

Finite-order translation kernels have a substantially wider range than the infinite-order class. Khare’s characterization of measurable TN_p functions for $p \geq 3$ provides a general finite-order setting [4]. The present proof instead derives a criterion specific to order four, using successive quotient derivatives and a curvature transport identity.

The theta normalization used here is the classical one in de Bruijn’s study of the Riemann Fourier kernel [5]. For the same kernel, Dimitrov and Xu relate Wronskians of Fourier and Laplace transforms to correlation kernels and real-zero criteria [6], while Csordas and Varga obtain strict concavity and moment inequalities connected with the Riemann Hypothesis [7]. Grochenig explains the distinct Schoenberg correspondence relevant to the Riemann Hypothesis: the PF function there is recovered from the reciprocal of a Laguerre–Polya entire function, rather than identified with the original Riemann kernel [8].

A preliminary version of the present work containing a PF_5 counterexample was sent to a publisher in July 2025. The order-five argument below uses an exact origin-confluent sign. The classification proved here is global strict PF_4 , hence exact order four; Theorem 1.3 separately gives a continuous finite-order Fourier separator.

3 The kernel and smooth even continuation

Put

$$\vartheta(x) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 x}, \quad \Theta(t) = \vartheta(e^{2t}), \quad H(t) = e^{t/2} \Theta(t).$$

The theta transformation $\vartheta(x) = x^{-1/2} \vartheta(1/x)$ gives

$$\Theta(-t) = e^t \Theta(t), \quad H(-t) = H(t).$$

The series and all its derivatives converge locally uniformly, so H is real analytic and even. Define

$$\Phi(t) = \frac{1}{2} \left(H''(t) - \frac{1}{4} H(t) \right).$$

Termwise differentiation gives, for every real t ,

$$\Phi(t) = \sum_{n \geq 1} \left(4\pi^2 n^4 e^{9t/2} - 6\pi n^2 e^{5t/2} \right) e^{-\pi n^2 e^{2t}}. \quad (3.1)$$

Thus the positive-side formula often written with $|t|$ is the restriction of a real-analytic even function. In particular, smoothness at the origin is not inferred from numerical coverage.

For $t \geq 0$, the coefficient of the n -th exponential can be factored as

$$2\pi n^2 e^{5t/2} (2\pi n^2 e^{2t} - 3) > 0,$$

because $2\pi > 3$. Hence every term in (3.1) is positive and $\Phi(t) > 0$ on $[0, \infty)$. Evenness gives $\Phi(t) > 0$ on all of $\mathbb{R}$, so the following logarithm is globally defined.

Write

$$\ell = \log \Phi, \quad s = \ell', \quad q = -s' = -\ell''.$$

For $a < b$, define

$$A(a, b) = s(a) - s(b), \quad M(a, b) = \frac{q(b) - q(a)}{A(a, b)}. \quad (3.2)$$

When $q > 0$, s is strictly decreasing and $A(a, b) > 0$.

4 Certified one-variable inputs

Define

$$F_1 = qq'' - (q')^2, \quad F_2 = q^3 - F_1.$$

For the logarithmic cumulants $c_j = \ell^{(j)}$, set

$$\begin{aligned} m_0 &= 1, & m_1 &= 0, & m_2 &= c_2, & m_3 &= c_3, \\ m_4 &= c_4 + 3c_2^2, & m_5 &= c_5 + 10c_2c_3, \\ m_6 &= c_6 + 15c_2c_4 + 10c_3^2 + 15c_2^3, \end{aligned}$$

and

$$\mathcal{C}_4(t) = \det[m_{i+j-2}(t)]_{i,j=1}^4. \quad (4.1)$$

Lemma 4.1 (Certified input). *For the kernel Φ ,*

$$q(t) > 0, \quad F_2(t) > 0, \quad \mathcal{C}_4(t) > 0 \quad (t \in \mathbb{R}).$$

The proof combines exact rational polynomial inequalities with analytic theta-tail bounds. For $u \geq 0$, put

$$x = \pi e^{2u}, \quad y = e^{-3x}, \quad P_0(z) = 2z - 3,$$

and recursively

$$P_{j+1}(z) = \left(\frac{5}{2} - 2z\right) P_j(z) + 2zP'_j(z).$$

After division by the positive first theta mode, the first two modes give the normalized jet entries

$$a_j(x, y) = \frac{P_j(x) + 4yP_j(4x)}{P_0(x)}.$$

Exact rational Bernstein coefficients on three fixed algebraic domains prove

$$q_{1,2} > 10, \quad (F_2)_{1,2} > 1000, \quad (\mathcal{C}_4)_{1,2} > 50,000,000.$$

The complete contribution of modes $n \geq 3$ is then bounded by one monotone polynomial-geometric estimate. Its changes in the cleared sign numerators are smaller than the three displayed margins, both on $\pi \leq x \leq 5$ and on $x \geq 5$. Evenness completes the real line. All inequalities, including the derivative envelopes used on the tail, are derived in Appendix B. The exact replay uses no floating point, no interval subdivision, and no FLINT or Arb arithmetic.

5 Order three and the positive slope functional

For $\xi < m < r$, define

$$\Lambda(\xi; m, r) = A(\xi, r) + M(\xi, m) - M(m, r). \quad (5.1)$$

We give the full estimate used to prove its positivity.

Set $f_0 = q'/q$. Because $s' = -q$,

$$M(a, b) = \frac{\int_a^b q'(t) dt}{\int_a^b q(t) dt} = \frac{\int_a^b f_0(t) q(t) dt}{\int_a^b q(t) dt};$$

thus $M(a, b)$ is a q -weighted mean of f_0 . Choose a point $u \in [\xi, m]$ where f_0 is minimal on the left interval and a point $v \in [m, r]$ where it is maximal on the right. Then

$$M(m, r) - M(\xi, m) \leq f_0(v) - f_0(u) \leq \int_{\xi}^r \max(f'_0(t), 0) dt.$$

Since

$$f'_0 = \frac{qq'' - (q')^2}{q^2} = \frac{F_1}{q^2}$$

and $A(\xi, r) = \int_{\xi}^r q(t) dt$, equation (5.1) yields

$$\Lambda(\xi; m, r) \geq \int_{\xi}^r \left(q - \frac{\max(F_1, 0)}{q^2} \right) dt \quad (5.2)$$

$$= \int_{\xi}^r \frac{q^3 - \max(F_1, 0)}{q^2} dt = \int_{\xi}^r \frac{\min(q^3, F_2)}{q^2} dt > 0. \quad (5.3)$$

The last inequality follows from Lemma 4.1 and $q > 0$.

The same quantity is the logarithmic slope derivative in the normalized order-three determinant. Hence (5.3), together with strict log-concavity, proves strict global PF₃. We rederive the needed Wronskian form in the next section because its signs also control order four.

6 Quotient-Wronskian reduction

Fix $y_1 < y_2 < y_3 < y_4$, put $u_j(t) = \Phi(t - y_j)$, and write $p_j = t - y_j$, so $p_4 < p_3 < p_2 < p_1$. Define

$$v_j = (u_j/u_1)', \quad w_j = (v_j/v_2)'.$$

Successive column division and differentiation give

$$W(u_1, u_2, u_3) = u_1^3 W(v_2, v_3), \quad (6.1)$$

$$W(u_1, u_2, u_3, u_4) = u_1^4 v_2^3 w_3^2 \frac{d}{dt} \left(\frac{w_4}{w_3} \right). \quad (6.2)$$

Let $\mathcal{T}$ denote simultaneous translation of all point arguments. From (3.2),

$$\mathcal{T}A(a, b) = q(b) - q(a), \quad \mathcal{T} \log A(a, b) = M(a, b). \quad (6.3)$$

The sign convention in (6.3) will be used below. Since

$$\frac{v_j}{v_2} = \frac{u_j}{u_2} \frac{A(p_j, p_1)}{A(p_2, p_1)},$$

direct differentiation gives

$$g_j := \mathcal{T} \log(v_j/v_2) = A(p_j, p_2) + M(p_j, p_1) - M(p_2, p_1). \quad (6.4)$$

The exact weighted-mean split

$$A(p_j, p_1)M(p_j, p_1) = A(p_j, p_2)M(p_j, p_2) + A(p_2, p_1)M(p_2, p_1)$$

turns (6.4) into

$$g_j = \frac{A(p_j, p_2)}{A(p_j, p_1)} \Lambda(p_j; p_2, p_1) > 0. \quad (6.5)$$

Thus $w_j = (v_j/v_2)g_j > 0$, and (6.1) has the strict order-three sign.

Define

$$\Psi(\xi; m, r) = \Lambda(\xi; m, r) + \mathcal{T} \log \Lambda(\xi; m, r).$$

Because $w_j = (v_j/v_2)g_j$,

$$\mathcal{T} \log(w_4/w_3) = (g_4 + \mathcal{T} \log g_4) - (g_3 + \mathcal{T} \log g_3). \quad (6.6)$$

Equation (6.5) and (6.3) give

$$\mathcal{T} \log g_j = M(p_j, p_2) - M(p_j, p_1) + \mathcal{T} \log \Lambda_j,$$

where $\Lambda_j = \Lambda(p_j; p_2, p_1)$. A second use of the same mean split gives

$$g_j + M(p_j, p_2) - M(p_j, p_1) = \Lambda_j - A(p_2, p_1).$$

The common last term cancels in (6.6), yielding the exact identity

$$\mathcal{T} \log(w_4/w_3) = \Psi(p_4; p_2, p_1) - \Psi(p_3; p_2, p_1). \quad (6.7)$$

Lemma 6.1 (Iterated quotient integral). *Let $f_1, \dots, f_k$ be continuously differentiable on an interval, let $f_1 > 0$, and put $G_j = f_j/f_1$. For $t_1 < \dots < t_k$,*

$$\det[f_j(t_i)]_{i,j=1}^k = \prod_{i=1}^k f_1(t_i) \int_{t_1}^{t_2} \cdots \int_{t_{k-1}}^{t_k} \det[G'_{j+1}(\tau_i)]_{i,j=1}^{k-1} d\tau_1 \cdots d\tau_{k-1}. \quad (6.8)$$

The identity may be applied repeatedly whenever the first member of the new derivative family is positive.

Proof. Factor $f_1(t_i)$ from row i . The first column is then one. Replace successive rows, from bottom to top, by their difference with the preceding row. Each remaining entry is $G_j(t_{i+1}) - G_j(t_i) = \int_{t_i}^{t_{i+1}} G'_j(\tau) d\tau$. Determinant multilinearity gives (6.8). Since $\tau_i \in [t_i, t_{i+1}]$, the integration nodes remain ordered, so the same argument applies to the derivative determinant. $\square$

Proposition 6.2 (Three-point equivalence). *Under the strict lower-order positivity already proved, global PF_4 is equivalent to*

$$\partial_\xi \Psi(\xi; m, r) \leq 0 \quad (\xi < m < r).$$

Strict inequality implies strict global PF_4 .

Proof. If $\partial_\xi \Psi \leq 0$, then $p_4 < p_3$ makes the right side of (6.7) nonnegative. All prefactors in (6.2) are positive. The Wronskian is therefore nonnegative, and is positive in the strict case.

For clarity, applying Lemma 6.1 three times to the order-four minor gives, with $I_i = [t_i, t_{i+1}]$,

$$\begin{aligned} \det[u_j(t_i)]_{i,j=1}^4 &= \prod_{i=1}^4 u_1(t_i) \int_{I_1 \times I_2 \times I_3} \prod_{i=1}^3 v_2(\tau_i) \\ &\quad \times \int_{\tau_1}^{\tau_2} \int_{\tau_2}^{\tau_3} w_3(\sigma_1) w_3(\sigma_2) \int_{\sigma_1}^{\sigma_2} \left(\frac{w_4}{w_3} \right)'(\rho) d\rho d\sigma_2 d\sigma_1 d\tau. \end{aligned}$$

Every factor is positive in the strict case, and every integration interval has positive length. This proves positivity of every ordered collocation minor, not only of the confluent Wronskian.

Conversely, assume global PF_4 . Let all four row nodes coalesce, divide by their positive Vandermonde, and pass to the limit. The resulting translate Wronskian is nonnegative. Equations (6.2) and (6.7) imply $\Psi(p_4; p_2, p_1) \geq \Psi(p_3; p_2, p_1)$ whenever $p_4 < p_3$. Thus $\Psi(\cdot; m, r)$ is nonincreasing, which is equivalent to the stated differential inequality because it is smooth. $\square$

7 Curvature coordinate and sign bridge

Since $q > 0$,

$$y(t) = -s(t), \quad Q(y(t)) = q(t)$$

defines a global strictly increasing coordinate on its image, with $dy/dt = q(t) = Q(y) > 0$. Primes in this and the next three sections mean y -derivatives. Put

$$\rho = 1 - Q'' = \frac{F_2}{q^3} > 0, \quad \kappa = 2 - Q'' = 1 + \rho > 1.$$

For $\xi < m < r$, write

$$p = y(\xi), \quad z = y(m), \quad w = y(r), \quad L = z - p, \quad R = w - z.$$

Because $A(a, b) = y(b) - y(a)$ and $M(a, b) = Q[y(a), y(b)]$,

$$\Lambda = (w - p) + Q[p, z] - Q[z, w]. \quad (7.1)$$

Two integrations of $\kappa = 2 - Q''$ give

$$\Lambda = \frac{1}{L} \int_p^z (y - p) \kappa(y) dy + \frac{1}{R} \int_z^w (w - y) \kappa(y) dy, \quad (7.2)$$

$$\delta := -\partial_p \Lambda = \frac{1}{L^2} \int_p^z (z - y) \kappa(y) dy > 0. \quad (7.3)$$

Simultaneous translation becomes

$$\mathcal{T} = Q(p) \partial_p + Q(z) \partial_z + Q(w) \partial_w,$$

and $\partial_\xi = Q(p) \partial_p$. Thus $\Lambda_\xi = -Q(p) \delta$.

Define

$$\mathcal{N} = \delta \Lambda^2 + Q'(p) \delta \Lambda + \Lambda \mathcal{T} \delta - \delta \mathcal{T} \Lambda. \quad (7.4)$$

Differentiating $\Psi = \Lambda + \mathcal{T} \log \Lambda$, commuting simultaneous translation with ∂_ξ , and collecting over Λ^2 gives

$$\partial_\xi \Psi = -\frac{Q(p)}{\Lambda^2} \mathcal{N}. \quad (7.5)$$

Equivalently,

$$\frac{\mathcal{N}}{\delta \Lambda} = K := \Lambda + Q'(p) + \mathcal{T} \log \delta - \mathcal{T} \log \Lambda. \quad (7.6)$$

It remains to prove $K > 0$.

8 The confluent invariant in curvature form

Define

$$D = 3(\kappa - 1) - \{Q(\log \kappa)'\}' \quad (8.1)$$

Lemma 8.1 (Confluent-to-curvature identity). *The determinant (4.1) satisfies*

$$\mathcal{C}_4 = Q^6 \kappa^2 D. \quad (8.2)$$

Consequently $D > 0$ on $\mathbb{R}$.

Proof. The change of variable gives $d/dt = Q d/dy$ and $c_2 = -Q$. Recursively applying this operator produces the five cumulants listed in Appendix A. Substitution into the Schur-complement form of (4.1) factors out Q^6 . The remaining factor is

$$\kappa^2 (3(\kappa - 1) - \{Q(\log \kappa)'\}'),$$

as shown by the common expanded numerator in Appendix A. This proves (8.2) without a numerical assumption. Lemma 4.1, $Q > 0$, and $\kappa > 0$ then imply $D > 0$. $\square$

The determinant definition is primary: it is the doubly confluent fourth-order translation minor after division by the positive row and column Vandermonde factors and by Φ^4 . The explicit thirteen-term expansion is included in Appendix A for independent checking of the certificate input.

9 Exact transport identity

Normalize the triangular weights in (7.2) and (7.3) as probability measures:

$$d\mu(y) = \frac{(z-y)\kappa(y)}{L^2\delta} \mathbf{1}_{[p,z]}(y) dy, \quad (9.1)$$

$$d\nu(y) = \frac{(y-p)\kappa(y)}{L\Lambda} \mathbf{1}_{[p,z]}(y) dy + \frac{(w-y)\kappa(y)}{R\Lambda} \mathbf{1}_{[z,w]}(y) dy. \quad (9.2)$$

Set

$$A_0(y) = 3(y - Q'(y)) - Q(y) \frac{\kappa'(y)}{\kappa(y)}.$$

Then $A'_0 = D$.

Lemma 9.1 (Transport cancellation). *With K as in (7.6),*

$$K = \mathbb{E}_\nu[A_0] - \mathbb{E}_\mu[A_0]. \quad (9.3)$$

Proof. Two elementary primitives carry the complete cancellation:

$$\kappa A_0 = H', \quad H = 3y^2 - 3Q - 3yQ' + (Q')^2 + QQ'', \quad (9.4)$$

$$H = J', \quad J = y^3 - 3yQ + QQ'. \quad (9.5)$$

Both identities follow by one differentiation.

Let $I_L = J(z) - J(p)$ and $I_R = J(w) - J(z)$. Integrating the linear weights in (9.1)–(9.2) by parts once gives

$$\mathbb{E}_\nu[A_0] - \mathbb{E}_\mu[A_0] = \frac{-I_L/L + I_R/R}{\Lambda} + \frac{LH(p) - I_L}{L^2\delta}. \quad (9.6)$$

On the other hand, differentiation of the triangular integrals along $\mathcal{T}$ gives

$$\mathcal{T}\Lambda = U(z, w) - U(p, z), \quad \mathcal{T}\delta = \frac{U(p, z) - (Q(z) - Q(p))\delta - Q(p)\kappa(p)}{L},$$

where

$$U(c, d) = Q(c) + Q(d) - \frac{Q(d)Q'(d) - Q(c)Q'(c)}{d - c} + \frac{(Q(d) - Q(c))^2}{(d - c)^2}. \quad (9.7)$$

Substitution in (7.6) yields

$$K = \Lambda + Q'(p) - Q[p, z] + \frac{U(p, z) - U(z, w)}{\Lambda} + \frac{U(p, z) - Q(p)\kappa(p)}{L\delta}. \quad (9.8)$$

The endpoint identity equating (9.6) and (9.8), with every substitution exposed, is Appendix C. It is an identity in the endpoint positions and the values of Q, Q', Q'' ; no bound or polynomial model is used. This proves (9.3). $\square$

10 The positive crossing kernel

Let F_μ, F_ν be the CDFs of (9.1)–(9.2), and put

$$W(y) = F_\mu(y) - F_\nu(y).$$

Lemma 10.1 (Strict stochastic ordering). *One has $W(y) > 0$ for $p < y < w$, while $W(p) = W(w) = 0$.*

Proof. On $[z, w]$, $F_\mu = 1$, so $W = 1 - F_\nu > 0$ before w . On (p, z) , the ratio of the two positive densities is

$$R_0(y) = \frac{d\mu/dy}{d\nu/dy} = \frac{\Lambda(z-y)}{L\delta(y-p)}. \quad (10.1)$$

Writing $C = \Lambda/(L\delta) > 0$, direct differentiation gives

$$R'_0(y) = -\frac{C(z-p)}{(y-p)^2} < 0. \quad (10.2)$$

Thus R_0 decreases strictly from $+\infty$ to 0. Hence $W' = \mu' - \nu'$ changes sign exactly once, from positive to negative. Since $W(p) = 0$ and

$$W(z) = 1 - F_\nu(z) = \nu((z, w]) > 0,$$

the function cannot return to zero on $(p, z]$. The right interval was already handled. $\square$

Integration by parts for CDFs, using $A'_0 = D$ and the common total mass, turns Lemma 9.1 into

$$K = \int_p^w W(y)D(y)dy. \quad (10.3)$$

Combining (7.6), (8.2), and (10.3) gives the central identity

$$\mathcal{N} = \delta\Lambda \int_p^w W(y) \frac{\mathcal{C}_4(y)}{Q(y)^6 \kappa(y)^2} dy > 0. \quad (10.4)$$

Every factor in the interior is strictly positive.

11 Completion and exact order

Equation (7.5) and (10.4) imply

$$\partial_\xi \Psi(\xi; m, r) < 0 \quad (\xi < m < r).$$

Proposition 6.2 therefore proves Theorem 1.1.

Michałowski also studied order-five Toeplitz obstructions numerically [9]; here a direct origin-confluent determinant suffices.

We now prove directly that the classification stops at order four. Put

$$H_k = \det[\Phi^{(i+j)}(0)]_{i,j=0}^{k-1}, \quad s_j = \Phi^{(j)}(0).$$

Since Φ is even, simultaneous parity permutations of the rows and columns give

$$H_4 = AB, \quad H_5 = CB,$$

where

$$A = s_0 s_4 - s_2^2, \quad B = s_2 s_6 - s_4^2,$$

and

$$C = \det \begin{pmatrix} s_0 & s_2 & s_4 \\ s_2 & s_4 & s_6 \\ s_4 & s_6 & s_8 \end{pmatrix}.$$

Lemma 11.1. *For the Riemann kernel,*

$$445 < A < 446, \quad 189223 < B < 189228, \quad -674000000 < C < -614000000.$$

In particular, $H_4 > 0$ and $H_5 < 0$.

Proof. For $x = \pi n^2 e^{2u}$, the n -th summand of $\Phi(u)$ is $e^{u/2-x} P_0(x)$, where $P_0(x) = 2x(2x-3)$. If its j -th derivative is $e^{u/2-x} P_j(x)$, then

$$P_{j+1} = 2xP'_j + (1/2 - 2x)P_j.$$

Thus

$$s_j = \sum_{n \geq 1} e^{-\pi n^2} P_j(\pi n^2).$$

The first three modes are enclosed by rational alternating-series bounds. Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ supplies rational bounds for π , and argument division followed by the degree-29 and degree-30 alternating Taylor sums supplies rational bounds for each exponential.

For the omitted modes, let S_j be the sum of the absolute coefficients of P_j . Since $\deg P_j = j+2$, $3 < \pi < 22/7$, and $e^3 > \sum_{r=0}^8 3^r/r! > 20$,

$$\sum_{n \geq 4} |P_j(\pi n^2)| e^{-\pi n^2} \leq \frac{S_j (22/7)^{j+2} 4^{2j+4} 20^{-16}}{1 - (5/4)^{20} 20^{-9}} \quad (j = 0, 2, 4, 6, 8).$$

All endpoints lie on $10^{-36}\mathbb{Z}$, and every arithmetic operation is rounded outward. Substitution into the displayed formulas gives the entirely rational sign chain

$$\begin{aligned} 0 < 445 < A < 446, \quad 0 < 189223 < B < 189228, \\ -674000000 < C < -614000000 < 0. \end{aligned}$$

Thus $AB > 0$ and $CB < 0$ without a floating-point determinant. The full grid endpoints and the standard-library verifier are included in the reproducibility archive. $\square$

Theorem 11.2. *The Riemann kernel is not PF_5 .*

Proof. For a C^{2k-2} function f , repeated row and column divided differences give, for increasing fixed tuples a, b ,

$$\lim_{h \downarrow 0} \frac{\det[f(z + h(a_i - b_j))]_{i,j=0}^{k-1}}{h^{k(k-1)} V(a) V(b)} = \frac{(-1)^{k(k-1)/2}}{\prod_{r=0}^{k-1} (r!)^2} \det[f^{(i+j)}(z)]_{i,j=0}^{k-1}.$$

Take $k = 5$, $z = 0$, and $a_i = b_i = i$. The sign factor is one, both Vandermonde factors are positive, and Lemma 11.1 makes the limit strictly negative. Hence $\det[\Phi((i-j)h)]_{i,j=0}^4 < 0$ for every sufficiently small positive h . These are translation minors at strictly increasing nodes. $\square$

Theorem 1.1 and Theorem 11.2 prove Corollary 1.2.

Remark 11.3 (Riemann Hypothesis boundary). Schoenberg's transform theorem says that the bilateral Laplace transform of a PF_∞ function is the reciprocal of a Laguerre–Polya entire function on its strip of convergence [1, 3]. The Fourier transform of Φ , up to the standard normalization, is the Riemann Ξ -function, which has known real zeros; hence Φ is unconditionally not PF_∞ . The Schoenberg formulation related to the Riemann Hypothesis instead asks for the inverse transform of the reciprocal of Ξ to be a Polya-frequency function [8]. Thus the finite-order classification proved here is a different statement. The next section further shows that strict PF_4 alone does not imply Fourier real-rootedness.

12 An explicit continuous PF_4 separator

We now prove Theorem 1.3. Put

$$c = \frac{1}{64}, \quad B(w) = \sum_{k=-3}^3 b_k w^k, \quad (b_{-3}, \dots, b_3) = (2, 10, 23, 30, 23, 10, 2),$$

and define

$$f_*(x) = e^{-cx^2} B(e^{2cx}) = \sum_{k=-3}^3 b_k e^{ck^2} e^{-c(x-k)^2}. \quad (12.1)$$

The Gaussian-mixture expression proves positivity and Schwartz decay; palindromy of (b_k) proves evenness.

12.1 Strict PF_3

For fixed x , give $K \in \{-3, \dots, 3\}$ probability proportional to $b_k e^{2ckx}$, and write κ_j for its cumulants. Differentiation of the finite log-partition function gives

$$q = 2c - (2c)^2 \kappa_2, \quad q' = -(2c)^3 \kappa_3, \quad q'' = -(2c)^4 \kappa_4. \quad (12.2)$$

If $\mu = \mathbb{E}K$, then $0 \leq \mathbb{E}[(K+3)(3-K)] = 9 - \kappa_2 - \mu^2$. Thus $0 \leq \kappa_2 \leq 9$, and hence

$$q \geq 2c(1 - 18c) = 2c \frac{23}{32} > 0. \quad (12.3)$$

Moreover,

$$\begin{aligned} F_2 &= q^3 - \{qq'' - (q')^2\} \\ &= q^3 + q(2c)^4 \kappa_4 + (2c)^6 \kappa_3^2. \end{aligned} \quad (12.4)$$

The identity $\kappa_4 = \mathbb{E}(K - \mathbb{E}K)^4 - 3\kappa_2^2$ gives $\kappa_4 \geq -243$. Dropping the final nonnegative term in (12.4) gives

$$F_2 \geq q\{q^2 - 3888c^4\}.$$

At $c = 1/64$, the two coefficients after division by c^2 are

$$q^2/c^2 \geq \frac{529}{256}, \quad 3888c^2 = \frac{243}{256}.$$

Thus $F_2 \geq qc^2(143/128) > 0$ globally. The weighted-mean criterion of Section 5 proves strict PF_3 .

12.2 Exact order-four density

Let $\epsilon = 2c = 1/32$, $t = e^{\epsilon x}$, and

$$P(t) = t^3 B(t) = 2 + 10t + 23t^2 + 30t^3 + 23t^4 + 10t^5 + 2t^6.$$

For $E = t \frac{d}{dt}$, define

$$N_1 = tP', \quad N_{j+1} = t(N_j'P - jN_jP'). \quad (12.5)$$

The quotient rule gives $E^j \log P = N_j/P^j$. Therefore, for $\ell_j = (\log f_*)^{(j)}$,

$$\ell_2 = \frac{-\epsilon P^2 + \epsilon^2 N_2}{P^2}, \quad \ell_j = \frac{\epsilon^j N_j}{P^j} \quad (3 \leq j \leq 6). \quad (12.6)$$

Insert these jets into the central-moment determinant (4.1). Every monomial has derivative weight twelve, so every term may first be put over the common denominator P^{12} . The resulting numerator has the exact factor P^8 . After cancellation,

$$\mathcal{C}_4[f_*](x) = \frac{\tilde{H}(t)}{2^{54}P(t)^4}. \quad (12.7)$$

The exact expansion printed in Appendix E proves that $\tilde{H}$ is palindromic of degree 24 and that all 25 integer coefficients are strictly positive. Hence $\mathcal{C}_4[f_*] > 0$ on $\mathbb{R}$.

The established inequalities $q > 0$ and $F_2 > 0$ also give

$$\kappa = 2 - Q'' = \frac{q^3 + F_2}{q^3} > 0.$$

The generic transport identity (10.4) now has strictly positive density and crossing weight for f_* . Thus $\partial_\xi \Psi < 0$, and Proposition 6.2 proves that f_* is strict PF_4 .

12.3 Nonreal Fourier zeros

With $\hat{f}(z) = \int_{\mathbb{R}} f(x)e^{-izx} dx$, Gaussian translation gives

$$\hat{f}_*(z) = \sqrt{\frac{\pi}{c}} e^{-z^2/(4c)} \sum_{k=-3}^3 b_k e^{ck^2} e^{-ikz}. \quad (12.8)$$

Put $w = e^{-iz}$ and $u = w + w^{-1}$. The Laurent factor in (12.8) vanishes exactly when

$$R_c(u) = 2e^{9c}u^3 + 10e^{4c}u^2 + (23e^c - 6e^{9c})u + 30 - 20e^{4c} \quad (12.9)$$

vanishes. At $c = 1/64$, the exact calculation in Appendix E gives $\text{disc}(R_c) < 0$. Thus the real cubic has a nonreal conjugate pair. For $|w| = 1$, however, $w + w^{-1}$ is real. The corresponding w -roots therefore lie off the unit circle and lift through $w = e^{-iz}$ to zeros with $\text{Im } z \neq 0$. The Gaussian factor is zero-free. This completes the proof of Theorem 1.3.

13 Exact verification and reproduction

The repository supplies five independent verifiers for the finite algebra and rational inequalities used above.

Object	Repository verifier	Arithmetic
Global $q, F_2, \mathcal{C}_4$	<code>scripts/verify_riemann_signs_exact.py</code>	rational polynomial
Quotient reduction	<code>scripts/verify_pf4_wronskian_reduction.py</code>	symbolic exact
Transport identity	<code>scripts/verify_pf4_transport_kernel.py</code>	symbolic exact
Origin order-five sign	<code>scripts/verify_pf5_origin.py</code>	rational enclosures
Continuous separator	<code>scripts/verify_continuous_pf4_separator.py</code>	rational polynomial

The global-sign verifier retains the first two theta modes exactly. It checks the 361 Bernstein and decay inequalities summarized in Appendix B, the 140 derivative-envelope coefficients, and the analytic geometric bounds for all modes $n \geq 3$. Its launcher selects SymPy's pure-Python rational domain and refuses a FLINT import. No floating-point comparison or interval subdivision decides a sign.

The order-five verifier is independent of that calculation. It uses integer arithmetic with outward rounding on $10^{-36}\mathbb{Z}$, Machin bounds for π , alternating Taylor bounds for three theta modes, and the geometric tail in Lemma 11.1. The separator verifier reconstructs the central determinant, checks the exact P^8 cancellation, verifies all 25 coefficients in (12.7), and reproduces the rational discriminant inequality in Appendix E. The generated coefficient record has SHA-256

d0f91259919c7f237ef65068041122656636ca990fe6ef41a146dd53eaa1dba9.

The quotient and transport scripts check identities for generic smooth jets. Their numerical evaluations at selected points are regression tests only; all sign conclusions follow from the displayed identities and exact inequalities.

From a clean checkout with Python 3.13, SymPy 1.14.0, and mpmath 1.3.0, the single replay command is

`python scripts/replay_paper.py.`

It prints five lines beginning PASS, one for each row above, followed by `status=paper evidence replay passed`. The SHA-256 of its complete standard output is

ec48df7f12724a948075be71d1b6771e8fb2ea735ab74fb490ab61eafcb7eea3.

Package versions and the complete artifact hashes are also recorded in `requirements-paper.txt` and `paper/RELEASE_MANIFEST.md`.

14 Declarations, availability, and computational provenance

Author contribution. Joshua Rainstar conceived and directed the project, developed the proof architecture, verified the exact computations, and prepared the manuscript.

Acknowledgments and computational assistance. The author acknowledges the use of OpenAI Codex and other disclosed AI systems for symbolic calculation, implementation, drafting, and review. The author takes responsibility for the mathematical claims and the final manuscript.

Conflict of interest. The author declares no competing interests.

Code and data availability. The public repository is <https://github.com/falseywinchnet/zeta>. The exact analytic proof was promoted in commit 0ebb95b141a1. The submission source, generated coefficient table, environment lock, replay tool, and PDF are frozen together under the versioned tag `pf4-paper-v1.0.0`; their SHA-256 values are listed in `paper/RELEASE_MANIFEST.md`. The complete evidence replay is `python scripts/replay_paper.py` from the repository root.

All theorem-bearing sign decisions are exact. Numerical values produced by the repository scripts are diagnostic only; there is no separate experimental dataset.

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A Confluent and curvature algebra

The central-moment Hankel matrix is

$$\begin{pmatrix} 1 & 0 & m_2 & m_3 \\ 0 & m_2 & m_3 & m_4 \\ m_2 & m_3 & m_4 & m_5 \\ m_3 & m_4 & m_5 & m_6 \end{pmatrix}.$$

Taking the Schur complement of the upper-left entry reduces its determinant to

$$\det \begin{pmatrix} m_2 & m_3 & m_4 \\ m_3 & m_4 - m_2^2 & m_5 - m_2 m_3 \\ m_4 & m_5 - m_2 m_3 & m_6 - m_3^2 \end{pmatrix}.$$

Substitution of the moments from Section 4 and ordinary three-by-three expansion gives

$$\mathcal{C}_4 = 12c_2^6 + 24c_2^4c_4 - 24c_2^3c_3^2 + 2c_2^3c_6 - 12c_2^2c_3c_5 + 7c_2^2c_4^2 \quad (\text{A.1})$$

$$+ 12c_2c_3^2c_4 + c_2c_4c_6 - c_2c_5^2 - 9c_3^4 - c_3^2c_6 + 2c_3c_4c_5 - c_4^3. \quad (\text{A.2})$$

In the curvature coordinate, repeated application of $Q d/dy$ to $c_2 = -Q$ gives

$$\begin{aligned} c_2 &= -Q, \\ c_3 &= -QQ', \\ c_4 &= -Q\{(Q')^2 + QQ''\}, \\ c_5 &= -Q\{(Q')^3 + 4QQ'Q'' + Q^2Q'''\}, \\ c_6 &= -Q\{(Q')^4 + 11Q(Q')^2Q'' + 4Q^2(Q'')^2 + 7Q^2Q'Q''' + Q^3Q''''\}. \end{aligned}$$

Substitution in the preceding polynomial gives

$$\mathcal{C}_4 = -Q^6\mathcal{P},$$

where

$$\begin{aligned} \mathcal{P} &= QQ''Q'''' - Q(Q''')^2 - 2QQ'''' + Q'Q''Q''' - 2Q'Q''' \\ &\quad + 3(Q'')^3 - 15(Q'')^2 + 24Q'' - 12. \end{aligned}$$

Now $\kappa = 2 - Q''$, and direct differentiation gives

$$\{Q(\log \kappa)'\}' = \frac{Q'\kappa' + Q\kappa''}{\kappa} - \frac{Q(\kappa')^2}{\kappa^2},$$

so, after multiplication by κ^2 and substitution of $\kappa' = -Q'''$, $\kappa'' = -Q''''$,

$$3(\kappa - 1) - \{Q(\log \kappa)'\}' = -\frac{\mathcal{P}}{(Q'' - 2)^2} = -\frac{\mathcal{P}}{\kappa^2}.$$

Therefore $\mathcal{C}_4 = Q^6\kappa^2D$, proving (8.2) by an exposed common numerator.

B Exact proof of the certified input

This appendix proves Lemma 4.1. The positive-side theta series has j -th derivative

$$2\pi n^2 e^{5u/2 - n^2 x} P_j(n^2 x), \quad x = \pi e^{2u},$$

where the polynomials P_j satisfy the recurrence in Section 4. Machin's identity and alternating series give $\pi > 157/50$; hence $P_0(x) = 2x - 3 > 0$.

B.1 Cleared sign polynomials

In addition to a_j , define the normalized remaining modes

$$e_j(x) = \sum_{n \geq 3} \frac{n^2 e^{-(n^2 - 1)x} P_j(n^2 x)}{P_0(x)}, \quad b_j = a_j + e_j.$$

For an indeterminate raw jet $(b_0, \dots, b_6)$, put

$$N_q = b_1^2 - b_0 b_2,$$

$$N_{F_2} = -b_0^4 b_2 b_4 + b_0^4 b_3^2 + b_0^3 b_1^2 b_4 - 2b_0^3 b_1 b_2 b_3 + 2b_0^3 b_2^3 \\ - 3b_0^2 b_1^2 b_2^2 + 3b_0 b_1^4 b_2 - b_1^6,$$

and

$$N_{C_4} = \det[b_{i+j}]_{i,j=0}^3.$$

Direct logarithmic differentiation and the central-moment determinant give

$$q = \frac{N_q}{b_0^2}, \quad F_2 = \frac{N_{F_2}}{b_0^6}, \quad C_4 = \frac{N_{C_4}}{b_0^4}. \quad (\text{B.1})$$

The thirteen-term cumulant expansion obtained from the last identity agrees with (A.1); this is an exact polynomial identity.

For orientation, retaining only the first mode gives

$$q_1 = \frac{4x(4x^2 - 12x + 15)}{(2x - 3)^2}, \\ (F_2)_1 = \frac{64x^3(64x^6 - 576x^5 + 2448x^4 - 6432x^3 + 10044x^2 - 8100x + 2295)}{(2x - 3)^6}, \\ (C_4)_1 = \frac{49152x^6(16x^4 - 96x^3 + 360x^2 - 840x + 945)}{(2x - 3)^4}.$$

After $x = 5 + z$, every numerator has positive coefficients. The second mode is retained exactly because it supplies the larger margin near the origin.

B.2 Exact two-mode margin

Alternating Taylor bounds for the exponential imply

$$0 < y < \frac{1}{12000} \quad (x \geq \pi),$$

and

$$\frac{1}{23000} < y < \frac{1}{12000} \quad \left(\frac{157}{50} \leq x \leq \frac{10}{3} \right).$$

We use the following elementary certificate lemma. If $p(s, t) = \sum p_{ij} s^i t^j$ has bidegree (d, e) , its Bernstein coefficients on the unit square are

$$\beta_{k\ell} = \sum_{i \leq k, j \leq \ell} p_{ij} \frac{\binom{k}{i}}{\binom{d}{i}} \frac{\binom{\ell}{j}}{\binom{e}{j}}.$$

The power-to-Bernstein identity proves that $\beta_{k\ell} > 0$ for every k, ℓ implies $p > 0$ on the square.

Apply this lemma after affine rational changes of variables. For q , one rectangle ends at $x = 10/3$, followed by a positive base polynomial minus exponentially decreasing corrections. For F_2 , dependent bounds on the first rectangle are followed by a rectangle through $x = 5$ and a positive half-strip expansion. For C_4 , the first rectangle is followed by two negative odd-power corrections whose ratios to the positive base decrease. The derivative of each ratio has a numerator with positive coefficients after the indicated shift. The 361 resulting rational coefficient inequalities give

$$q_{1,2} > 10, \quad (F_2)_{1,2} > 1000, \quad (C_4)_{1,2} > 50,000,000. \quad (\text{B.2})$$

This is a finite algebraic proof on fixed domains.

For reference, the complete finite certificate is summarized below. Write $\mathcal{A} = (2x-3)+4(8x-3)y$, which is positive on every listed domain. Rows Q_0, F_0, F_1, C_0 are Bernstein certificates after the indicated affine change to the unit square; the remaining rows use coefficient positivity after a half-line shift. In the two “ratio” rows, the listed fraction is the sum of the adverse coefficient ratios and is strictly less than one.

ID	Variables and domain	Degree	Positive denominator	Exact lower datum
Q_0	$157/50 \leq x \leq 10/3, 0 \leq y \leq 1/12000$	(4, 2)	$\mathcal{A}^2$	6613625187767/70312500000
Q_∞	$x = 10/3 + z, 0 \leq y \leq 1/12000$	(4, 2)	$\mathcal{A}^2$	ratio 47333/505500
F_0	$157/50 \leq x \leq 10/3, 1/23000 \leq y \leq 1/12000$	(12, 6)	$\mathcal{A}^6$	minimum > 1610047
F_1	$10/3 \leq x \leq 5, 0 \leq y \leq 1/22000$	(12, 6)	$\mathcal{A}^6$	29124973000/19683
F_∞	$x = 5 + z, y = v/3000000$	(12, 6)	$\mathcal{A}^6$	432/19073486328125
C_0	$157/50 \leq x \leq 10/3, 0 \leq y \leq 1/12000$	(14, 4)	$\mathcal{A}^4$	minimum > 9546672947
C_∞	$x = 10/3 + z, 0 \leq y \leq 1/12000$	(14, 4)	$\mathcal{A}^4$	ratio 65989337396066791/771657205500000000

These seven checks comprise 361 exact rational inequalities. The derivative envelopes used below contribute 140 further Bernstein coefficients, of bidegrees $(j+1, 1)$, $0 \leq j \leq 6$; their smallest coefficient is 44861/31250. Thus the compact domains and the differentiated tail majorants belong to the same exact certificate.

B.3 All later modes

For $t \geq 27$, shifting $t = 27 + z$ gives positive coefficients in $(-1)^j P_j(t)$ and in $2^{j+1}t^{j+1} - (-1)^j P_j(t)$. Therefore

$$0 < (-1)^j P_j(t) \leq 2^{j+1}t^{j+1}, \quad 0 \leq j \leq 6. \quad (\text{B.3})$$

The derivative numerator of the normalized $n = 3$ term is another positive shifted polynomial, so its absolute value decreases for $x \geq 3$. For $n \geq 4$, successive majorants have ratio at most

$$\left(\frac{5}{4}\right)^{16} e^{-27} < 10^{-9}.$$

Rational Taylor lower bounds for e^{24}, e^{27}, e^{45} then yield, on $\pi \leq x \leq 5$,

$$(|e_0|, \dots, |e_6|) < (7 \cdot 10^{-9}, 4 \cdot 10^{-7}, 2 \cdot 10^{-4}, 7 \cdot 10^{-4}, 3 \cdot 10^{-2}, 1.1, 41). \quad (\text{B.4})$$

The same Bernstein lemma gives

$$(|a_0|, \dots, |a_6|) < (2, 5, 20, 200, 2000, 60000, 1000000).$$

Expanding the three polynomials in (B.1) and using absolute coefficients gives

$$|\Delta N_q| < 0.000405, \quad |\Delta N_{F_2}| < 37.959, \quad |\Delta N_{C_4}| < 31,549,532. \quad (\text{B.5})$$

For $x \geq 5$, (B.3) gives

$$|a_j| \leq 2^{j+2}x^j, \quad |e_j| \leq 2^{j+2}3^{2j+4}x^j e^{-8x}.$$

The raw weights of N_q, N_{F_2}, N_{C_4} are respectively 2, 6, 12. Thus every perturbation majorant is a positive multiple of $x^w e^{-8x}$, or a faster-decaying term, with $w \leq 12$; it decreases for $x \geq 5$. A single rational Taylor bound at $x = 5$ gives

$$|\Delta N_q| < 6.49 \cdot 10^{-11}, \quad |\Delta N_{F_2}| < 0.03, \quad |\Delta N_{C_4}| < 418,287. \quad (\text{B.6})$$

To compare like quantities, note that

$$a_0 = 1 + 4y \frac{P_0(4x)}{P_0(x)} > 1,$$

and, by homogeneity,

$$N_{q,1,2} = q_{1,2}a_0^2, \quad N_{F_2,1,2} = (F_2)_{1,2}a_0^6, \quad N_{C_4,1,2} = (C_4)_{1,2}a_0^4.$$

Consequently (B.2) supplies the same respective lower margins 10, 1000, and 50,000,000 for the cleared numerators. Both perturbation bounds (B.5) and (B.6) are strictly smaller. Finally $b_0 = a_0 + e_0 > 0$, so (B.1) proves all three signs for $u \geq 0$. Evenness proves Lemma 4.1.

C Endpoint algebra for the transport cancellation

This appendix records the algebra suppressed by the phrase “simplifying the endpoint expression.” Put

$$z = p + L, \quad w = z + R, \quad a = \frac{Q(z) - Q(p)}{L}, \quad b = \frac{Q(w) - Q(z)}{R}.$$

Direct integration of $2 - Q''$ gives

$$\delta = \frac{L + Q'(p) - a}{L}, \quad \Lambda = L + R + a - b. \quad (\text{C.1})$$

Define

$$J_y = y^3 - 3yQ(y) + Q(y)Q'(y), \\ H_p = 3p^2 - 3Q(p) - 3pQ'(p) + Q'(p)^2 + Q(p)Q''(p),$$

and, for endpoint data,

$$U(c, d) = Q(c) + Q(d) - \frac{Q(d)Q'(d) - Q(c)Q'(c)}{d - c} + \frac{(Q(d) - Q(c))^2}{(d - c)^2}.$$

With $I_L = J_z - J_p$ and $I_R = J_w - J_z$, the identity needed in Lemma 9.1 is exactly

$$\frac{-(J_z - J_p)/L + (J_w - J_z)/R}{\Lambda} + \frac{LH_p - (J_z - J_p)}{L^2\delta} \\ = \Lambda + Q'(p) - a + \frac{U(p, z) - U(z, w)}{\Lambda} + \frac{U(p, z) - Q(p)(2 - Q''(p))}{L\delta}.$$

To check it, substitute (C.1), replace z by $p + L$ and w by $p + L + R$, and multiply by the common denominator $L^2R\delta\Lambda$. The terms containing $Q'(w)$, then $Q'(z)$, then $Q''(p)$ cancel pairwise; the remaining endpoint values and position terms cancel after inserting $a = (Q(z) - Q(p))/L$ and $b = (Q(w) - Q(z))/R$. No differential equation for Q is used, so the identity holds for arbitrary smooth Q .

D Certificate inventory

The finite and symbolic parts of the proof are recorded in the following plain-text source files. Each record states the exact claim, its inputs, and the corresponding verifier.

Claim	Proof record
Global $q, F_2, \mathcal{C}_4$, order three	<code>sources/riemann-signs-analytic-certificate.md</code>
Quotient and three-point equivalence	<code>sources/pf4-wronskian-reduction-certificate.md</code>
Transport identity and completion	<code>sources/pf4-transport-kernel-certificate.md</code>
Origin order-five obstruction	<code>sources/pf5-origin-certificate.md</code>
Continuous strict-PF ₄ Fourier separator	<code>sources/continuous-pf4-separator-certificate.md</code>

The maintained source map `paper/PAPER_MAP.md` identifies the manuscript section in which each record is used. The single replay command in Section 13 checks the five corresponding verifier outputs.

E Exact separator calculation

This appendix records the algebraic boundary behind (12.7). Starting from the cumulants $\ell_2, \dots, \ell_6$, the verifier reconstructs the central moments

$$\begin{aligned}
m_0 &= 1, & m_1 &= 0, & m_2 &= \ell_2, & m_3 &= \ell_3, \\
m_4 &= \ell_4 + 3\ell_2^2, & m_5 &= \ell_5 + 10\ell_3\ell_2, \\
m_6 &= \ell_6 + 15\ell_4\ell_2 + 10\ell_3^2 + 15\ell_2^3,
\end{aligned}$$

and then forms $\det[m_{i+j}]_{i,j=0}^3$ directly. This independently regenerates the thirteen-term $\mathcal{C}_4$ polynomial; it does not import the expansion used for the Riemann kernel.

The recurrence (12.5) is then applied over $\mathbb{Q}[t]$. Substitution of (12.6) shows that each of the thirteen determinant monomials has common denominator P^{12} . After collecting, the numerator is divisible by P^8 in $\mathbb{Q}[t]$. Cancel this factor and clear the remaining powers of two, writing

$$\mathcal{C}_4[f_*](x) = \frac{\tilde{H}(t)}{2^{54}P(t)^4}.$$

Exact rational arithmetic verifies

$$\deg \tilde{H} = 24, \quad [t^j]\tilde{H} = [t^{24-j}]\tilde{H} > 0 \quad (0 \leq j \leq 24),$$

with smallest integer coefficient

$$\min_{0 \leq j \leq 24} [t^j]\tilde{H} = 3221225472.$$

Equivalently, the smallest coefficient before multiplication by 2^{54} is $3/2^{24}$. Write

$$\tilde{H}(t) = \sum_{j=0}^{24} a_j t^j, \quad a_j = a_{24-j}.$$

The 13 independent integer coefficients are:

j	a_j	j	a_j
0	3221225472	7	357704506872240
1	61435985920	8	701640591264080
2	575070935296	9	1172880923200640
3	3512452441920	10	1684288128586089
4	15702254784944	11	2088645133531440
5	54633824917920	12	2243243092164284
6	153607464071178		

This table is generated by `scripts/generate_separator_coefficients.py`. Its machine-readable companion `generated/separator-coefficients.json` has SHA-256 prefix `d0f91259919c7f23`; the complete hash is recorded in the release manifest. Coefficient positivity proves the sign on the full domain $t > 0$. Exact spot reconstructions at $t = 1/3, 1, 4$ independently compare the central determinant with $\tilde{H}/(2^{54}P^4)$.

For the Fourier step, the identities

$$w + w^{-1} = u, \quad w^2 + w^{-2} = u^2 - 2, \quad w^3 + w^{-3} = u^3 - 3u$$

give (12.9) directly. Put

$$r = e^{1/64}, \quad x = r^2 = e^{1/32}.$$

Exact expansion gives

$$\text{disc}(R_c) = 4r^{10}G(x),$$

where

$$\begin{aligned} G(x) = & 432x^{13} - 4968x^9 + 900x^8 + 16200x^6 + 19044x^5 \\ & - 72600x^4 + 20000x^3 + 62100x^2 - 54334x + 13225. \end{aligned}$$

Writing $y = x - 1$ gives

$$\begin{aligned} G(1+y) = & -1 - 10y - 12y^2 + 680y^3 + 11532y^4 + 96660y^5 \\ & + 365400y^6 + 569664y^7 + 512172y^8 + 303912y^9 \\ & + 123552y^{10} + 33696y^{11} + 5616y^{12} + 432y^{13}. \end{aligned}$$

Moreover $0 < y < 1/30$. Indeed,

$$e < 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{18} = \frac{49}{18} < \frac{11}{4},$$

where $1/18 = (1/24) \sum_{j \geq 0} 4^{-j}$ bounds the tail beginning with $1/4!$. The binomial theorem gives

$$\frac{11}{4} < 1 + \frac{32}{30} + \frac{\binom{32}{2}}{30^2} + \frac{\binom{32}{3}}{30^3} = \frac{1891}{675} < \left(\frac{31}{30}\right)^{32}.$$

The sum of the positive terms at $y = 1/30$ is exactly

$$\sum_{j=3}^{13} \frac{[y^j]G(1+y)}{30^j} = \frac{1621193195302591}{36905625000000000} < 1.$$

The two omitted terms $-10y - 12y^2$ are negative. Consequently

$$G(1+y) < -1 + \sum_{j=3}^{13} \frac{[y^j]G(1+y)}{30^j} < 0,$$

and hence $\text{disc}(R_c) < 0$ without numerical evaluation.